

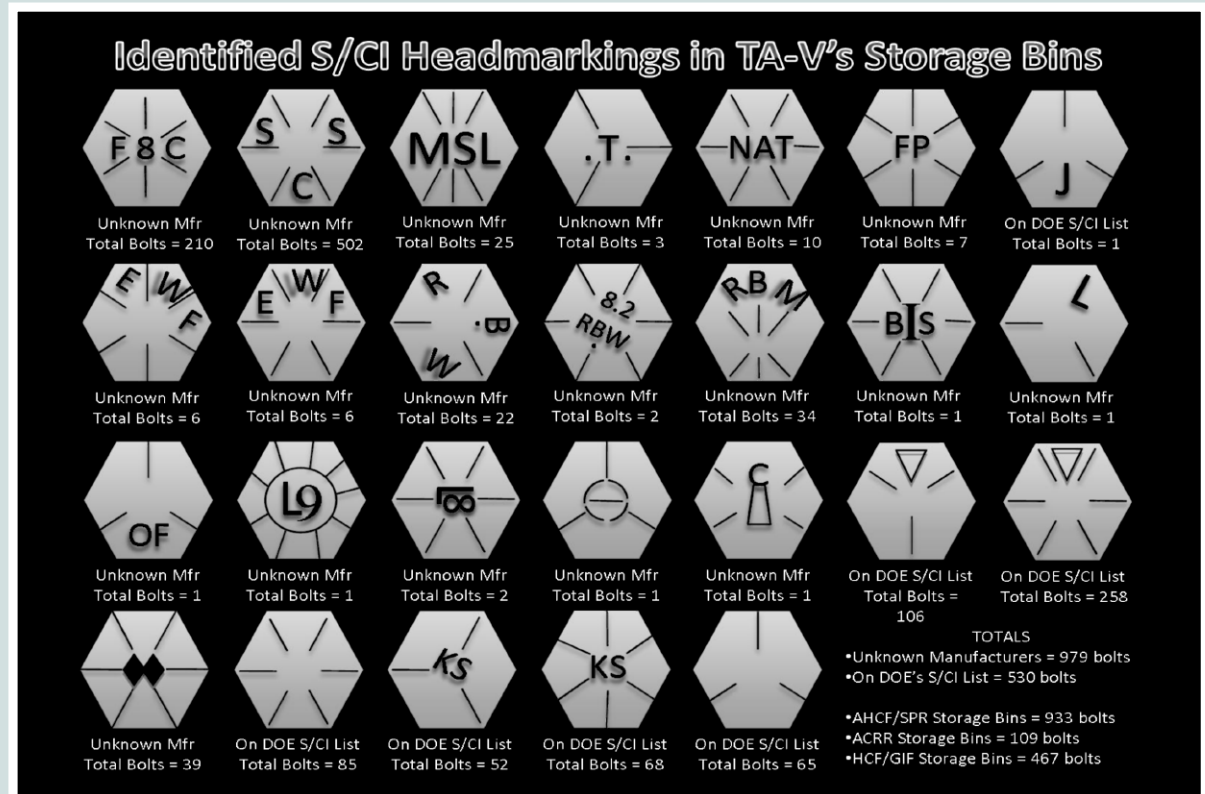


OPERATING EXPERIENCE SUMMARY

U.S. Department of Energy
Office of Health, Safety and Security
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Haste Makes Waste — Procurement Gone Wrong

1

On August 12, 2009, at Los Alamos National Laboratory (LANL), a facility operations director learned that incorrect parts for the fire alarm system had been delivered to the facility. Instead of the requested parts (fire pull stations), fire alarm control panels were ordered from the vendor, processed through the procurement group, and accepted by the receipt inspection group upon delivery to the site. (ORPS Report NA--LASO-LANL-CMR-2009-0009; final report issued January 21, 2010)

The LANL procurement cycle, from initiation of procurement through receipt inspection and delivery, normally takes 10 days. Procuring and delivering the parts needed for the fire alarm system needed to be completed within 7 days, so obtaining the fire protection pull stations was categorized as a Priority 1 (“rush status”). The Priority 1 category resulted in all of the steps leading to procurement of the pull stations being accomplished within 1 day.

The requestor learned that the preferred vendor for the parts (original supplier of the fire alarm system) was no longer on the supplier list, so he contacted another recommended vendor, confirmed they could supply the parts as an overnight order, and received a quote. In addition, the requestor generated a bill of materials in the Work Management System database. As can be seen in Table 1-1, the vendor quote did not match the correct bill of materials.

Once the vendor quote was received and bill of materials prepared, time constraints began to impact the procurement process. The requestor was the first to review and approve the procurement package. He told investigators that he saw that the model number was correct on the bill of materials, but did not notice that the supplier of the parts shown was not the manufacturer originally identified in the quote or that the part description was “fire alarm control panel,” instead of the requested “fire alarm pull station.” He also said that normally he would have followed up with a phone call and confirmation e-mail to the vendor that re-iterated the order specifications, but because of the urgent need for the parts he did not.

The discrepancies also went unnoticed during reviews performed by the Quality Assurance Subject Matter Expert (QASME), the Designated Procurement Representative (DPR), and the Buyer, all of whom stated that they performed limited reviews due to schedule pressures, relying on the initial review by the requestor.

	Vendor Quote	Bill of Materials
Manufacturer	Firelite	FCI
Model Number	MS-2 (correct number is MS-2E)	FCI MS-2
Item Description	Fire Alarm Control Panel	Fire Alarm Pull Station, Dual Action
Part Number	Not listed	FCI part # 100-0615

**Table 1-1. Comparison of the vendor quote
to the bill of materials**



The procurement process not only set in motion receipt of the wrong parts, but also resulted in a breakdown of the receipt inspection process. On the morning that the parts arrived at the receiving dock, the DPR sent an email to the Receipt Inspector asking for a priority receipt inspection because of the urgent need for the parts to be delivered to the facility. Included in the email was a Certificate of Conformance and tear sheets provided by the vendor. In an effort to facilitate inspection and delivery, the DPR did not cross-check the information on the documents and assumed that the description on both the certificate and the tear sheets was correct. However, the two discrepancies (i.e., manufacturer and part description) remained.

Receipt inspectors are to verify the following six items: Certificate of Conformance, damage, material description, part number, quantity, and vendor. When an inspector identifies a discrepancy, the process is to contact procurement staff, notify them of the discrepancy, and resolve the issue. In this case, however, despite noticing the discrepancies, the inspector decided he could accept the delivery without contacting the DPR because the parts were urgently needed. He entered the inspection data into the e-Procurement system, where the items were accepted and approved for delivery.

As a result of the expedited procurement, all of the steps, including delivery of the parts, were completed within the required 7 days. Unfortunately, when the parts arrived at the facility and were found to be unusable, they had to be returned to the vendor. The correct parts arrived approximately 6 weeks later.

Problems in the procurement process that led to delivery of the incorrect parts included the following.

- Neither the requestor nor the QASME, both of whom had responsibility for an in-depth review and approval of the procurement paperwork, noticed that the procurement specifications listed in the paperwork were incorrect. The requestor had two review opportunities before proceeding to the next step in the process, but still did not notice the errors.
- There was an assumption that the procurement package was accurate based on the requestor's review since he was the technical expert. This assumption carried through each step of the process through receipt inspection. The QASME, DPR, Buyer, and Receipt Inspector all deferred to the technical knowledge of the requestor and assumed he had performed an in-depth review. Thus, they incorrectly assumed that the technical aspects of the package were correct and did not follow procedures (i.e., perform their own in-depth reviews of the package).
- Emphasis on quickly procuring the parts using the Priority 1 procurement status created a heightened perception of urgency that resulted in steps that normally occurred being omitted.

Assumptions, coupled with schedule pressures, often lead to errors such as occurred in this event. The emphasis on schedule replaced the emphasis on doing the task properly. Had each of the reviewers involved in the purchasing task followed normal procedures when performing his/her task, the correct parts would have been ordered. Instead, in an effort to "rush" the process, the needed parts were delayed by nearly 6 weeks.



The following lessons learned identified by the facility provide good recommendations to follow. Personnel must stay focused at all times on the tasks being performed, paying special attention that the work is being performed per applicable policies and/or procedure. Personnel should utilize Human Performance Improvement (HPI) tools (e.g., self checking, procedural compliance, attention to detail, etc.) as part of their daily activities to identify and resolve issues.

KEYWORDS: Procurement errors, receipt inspection, incorrect parts, fire alarm pull station, fire alarm control panel

ISM CORE FUNCTION: Perform Work within Controls

Many Suspect/Counterfeit Bolts Found During Extent of Condition Review

2

On August 18, 2009, at Sandia National Laboratories, facility operators discovered that two bolts that were Suspect Counterfeit Items (S/CI) had been installed on a gauge associated with an air compressor in the facility. An additional 88 S/CI bolts were later discovered, unused, in a storage bin in the facility. All of the S/CI bolts had headmarks that were 3 slashes (normally indicating a grade 5 bolt) and had “KS” stamped on them. (ORPS Report NA--SS-SNL-1000-2009-0009; revision to final report issued October 2, 2009)

On November 15, 2009, as a follow-on to the initial assessment, management decided to continue an extent of condition review for all Tech Area V (TA-V) facility storage bins to look for additional S/CI fasteners. (ORPS Report NA--SS-SNL-1000-2010-0003; final report issued February 22, 2010)

As a result of the review of the TA-V spare hardware storage bins, a total of 1,509 bolts were identified as either being on the DOE S/CI list or having headmarks that did not match any certified manufacturers; thus, the bolts were suspect. Out of the 1,509 bolts, 933 bolts were from the Auxiliary Hot Cell Facility and Sandia Pulse Reactor storage bins; 109 bolts were from the Annular Core Research Reactor storage bins; and 467 bolts were from Hot Cell Facility and Gamma Irradiation Facility storage bins. Figure 2-1 shows the suspect bolts segregated by the various S/CI headmarks known to DOE and the headmarks of unknown manufacturers. The bolts varied in size and materials, and not all had the same headmarks.



Figure 2-1. S/CI bolts found in spare hardware bins at TA-V



Several references were used to compare the headmarks on the bolts against known, registered manufacturer insignias. If the headmark was not listed in any of the references reviewed, the bolt was identified as “Unknown Manufacturer” and collected. The suspect bolts were then compared to the “Suspect/Counterfeit Part Headmark List” developed by the U.S. Customs Service and adopted by DOE. This list, posted at <http://www.anl.gov/PRO/General%20Information/webforms/suspect.pdf>, is a recommended reference for identifying S/CI bolts. Although the list has not been updated recently, it is still relevant for use during receipt inspection of newly purchased bolts.

To capture the results of their review, Sandia created a listing of the various headmarks found during their extent of condition review. Figure 2-2 shows the various headmarks on the SC/I bolts identified during the review. The 1,509 segregated bolts will be processed through the Sandia suspect/counterfeit program, and an informal causal analysis will be used to identify a path forward.

Additional Information

The efforts demonstrated by Sandia are a good representation of the quality of the DOE S/CI program, which is strengthened by the collaborative efforts of program managers and coordinators and by various inspection entities at the field sites, ranging from receipt inspectors to Quality Assurance inspectors to the workers, themselves. A recent study of occurrence reports related to S/CI, performed by DOE’s Office of Corporate Safety Analysis, identified the following.

- The number of ORPS reports filed for S/CI discoveries has remained relatively constant over the 4-year period covered in the study.
- Workers regularly make the highest number of S/CI discoveries, and awareness training may be a factor.

- Bolts, shackles, and fasteners are the most frequently reported S/CI mentioned by name, but most of the S/CI reported is in other categories that are not specified as clearly.
- Suspect/counterfeit electronics are a growing concern.
- All Program Secretarial Offices and sites are represented in the ORPS data, indicating that there is awareness of the S/CI issue throughout the Complex.
- The Hanford Site reports the largest number of S/CI discoveries, followed by Pantex and Sandia National Laboratories.

This DOE study will be finalized and posted on DOE’s Health, Safety and Security website in the near future.

As mentioned above, DOE, along with all government entities, is facing an increased threat of counterfeit electronics, which is the subject of a U.S. Department of Commerce report, *Defense Industrial Base Assessment: Counterfeit Electronics*, released in January 2010. In June 2007, the Department of the Navy, Naval Air Systems Command, asked the Bureau of Industry and Security, Office of Technology Evaluation (OTE), to conduct an assessment of counterfeit electronics in the defense industry. The data revealed that 39 percent of these companies had encountered counterfeit electronics and that the numbers were increasing. The report indicated that demonstrated weaknesses in inventory management, procurement, recordkeeping, reporting, inspection, and communication across government and industry were exacerbating the problem. The OTE report can be obtained at http://www.bis.doc.gov/defenseindustrialbaseprograms/osies/defmarketresearchrpts/final_counterfeit_electronics_report.pdf.

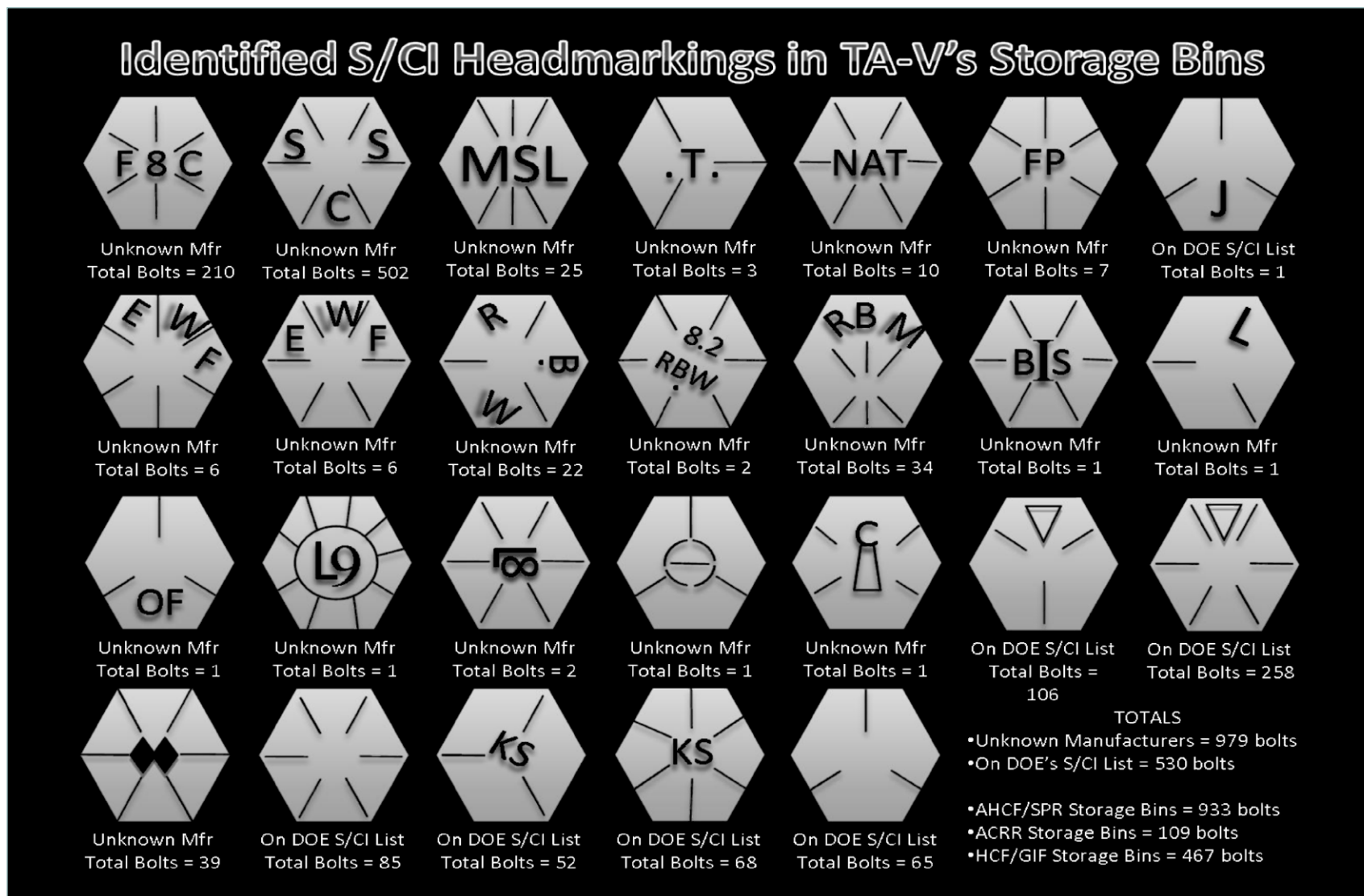


Figure 2-2. Graphic display of the headmarks on bolts found at Sandia



DOE is committed to ensuring that S/CI items and components are quickly identified and, if installed, are evaluated to ensure that they meet their intended functional and operability requirements. The DOE S/CI program is clearly defined in DOE Order 414.1C, Quality Assurance, and DOE G 414.1-3, Suspect/Counterfeit Items Guide, and is implemented through close cooperation between Headquarters and the various inspection entities at the field sites, ranging from receipt inspectors to Quality Assurance inspectors to the workers, themselves, who regularly make the highest number of S/CI discoveries.

KEYWORDS: Suspect, counterfeit, defective, headmark, electronics, bolts, S/CI

ISM CORE FUNCTION: Feedback and Improvement



OPERATING EXPERIENCE SUMMARY

The Office of Health, Safety and Security (HSS), Office of Analysis publishes the *Operating Experience Summary* to promote safety throughout the Department of Energy (DOE) complex by encouraging the exchange of lessons-learned information among DOE facilities.

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Commonly Used Acronyms and Initialisms

Agencies/Organizations	
ACGIH	American Conference of Governmental Industrial Hygienists
ANSI	American National Standards Institute
CPSC	Consumer Product Safety Commission
DOE	Department of Energy
DOT	Department of Transportation
EPA	Environmental Protection Agency
INPO	Institute for Nuclear Power Operations
NIOSH	National Institute for Occupational Safety and Health
NNSA	National Nuclear Security Administration
NRC	Nuclear Regulatory Commission
OSHA	Occupational Safety and Health Administration

Units of Measure	
AC	alternating current
DC	direct current
mg	milligram (1/1000th of a gram)
kg	kilogram (1000 grams)
psi (a)(d)(g)	pounds per square inch (absolute) (differential) (gauge)
RAD	Radiation Absorbed Dose
REM	Roentgen Equivalent Man
TWA	Time Weighted Average
v/kv	volt/kilovolt

Job Titles/Positions	
RCT	Radiological Control Technician

Authorization Basis/Documents	
JHA	Job Hazards Analysis
JSA	Job Safety Analysis
NOV	Notice of Violation
SAR	Safety Analysis Report
TSR	Technical Safety Requirement
USQ	Unreviewed Safety Question

Regulations/Acts	
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
CFR	Code of Federal Regulations
D&D	Decontamination and Decommissioning
DD&D	Decontamination, Decommissioning, and Dismantlement
RCRA	Resource Conservation and Recovery Act
TSCA	Toxic Substances Control Act

Miscellaneous	
ALARA	As low as reasonably achievable
HEPA	High Efficiency Particulate Air
HVAC	Heating, Ventilation, and Air Conditioning
ISM	Integrated Safety Management
MSDS	Material Safety Data Sheet
ORPS	Occurrence Reporting and Processing System
PPE	Personal Protective Equipment
QA/QC	Quality Assurance/Quality Control
SME	Subject Matter Expert